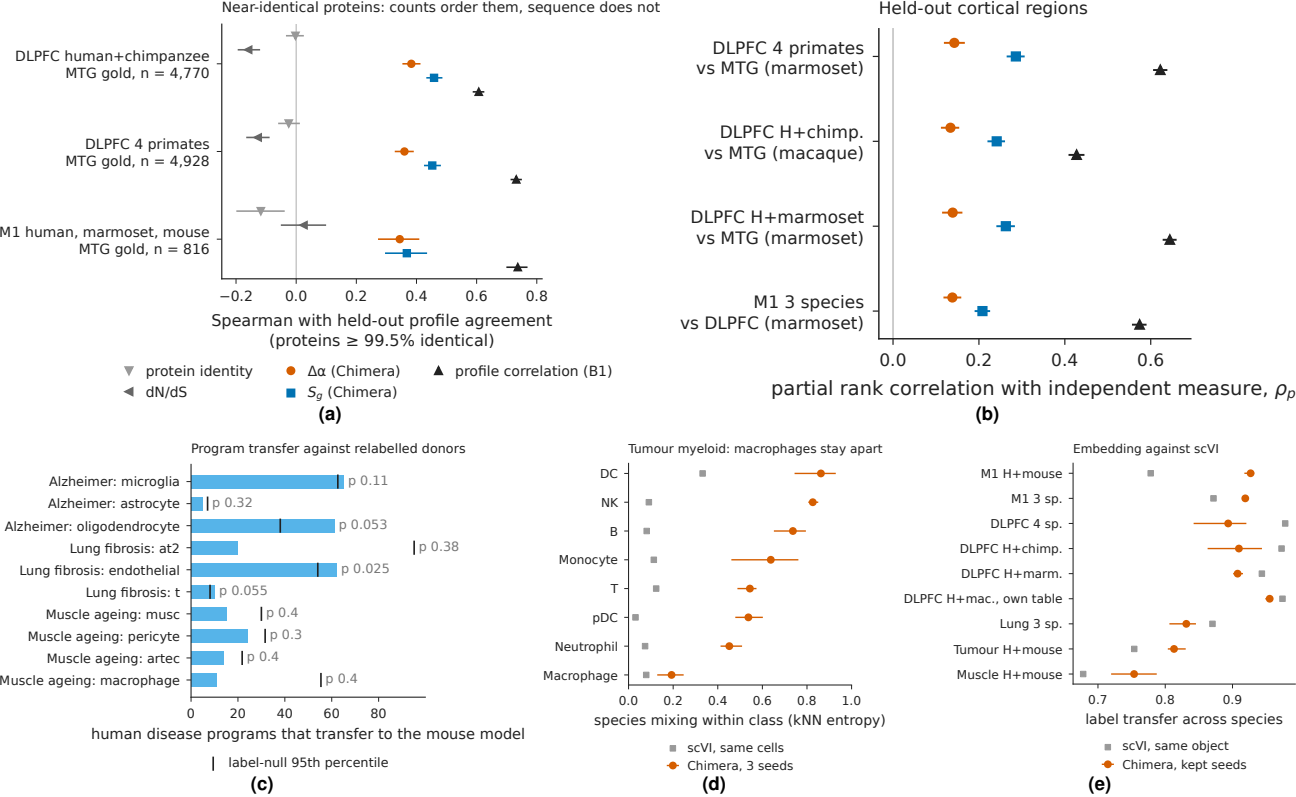

Near-identical proteins: counts order them, sequence does not
DLPFC human+chimpanzee
MTG gold, n = 4,770
DLPFC 4 primates
MTG gold, n = 4,928
M1 human, marmoset, mouse
MTG gold, n = 816
−0.2
0.0
0.2
0.4
0.6
0.8
Spearman with held-out profile agreement
(proteins ≥ 99.5% identical)
protein identity
dN/dS
Δα (Chimera)
S_g (Chimera)
profile correlation (B1)
(a)
Held-out cortical regions
DLPFC 4 primates
vs MTG (marmoset)
DLPFC H+chimp.
vs MTG (macaque)
DLPFC H+marmoset
vs MTG (marmoset)
M1 3 species
vs DLPFC (marmoset)
0.0
0.2
0.4
0.6
partial rank correlation with independent measure, ρ_p
(b)
Program transfer against relabelled donors
Alzheimer: microglia
Alzheimer: astrocyte
Alzheimer: oligodendrocyte
Lung fibrosis: at2
Lung fibrosis: endothelial
Lung fibrosis: t
Muscle ageing: musc
Muscle ageing: pericyte
Muscle ageing: artec
Muscle ageing: macrophage
p 0.11
p 0.32
p 0.053
p 0.38
p 0.025
p 0.055
p 0.4
p 0.3
p 0.4
p 0.4
0
20
40
60
80
human disease programs that transfer to the mouse model
label-null 95th percentile
(c)
Tumour myeloid: macrophages stay apart
DC
NK
B
Monocyte
T
pDC
Neutrophil
Macrophage
0.0
0.2
0.4
0.6
0.8
1.0
species mixing within class (kNN entropy)
scVI, same cells
Chimera, 3 seeds
(d)
Embedding against scVI
M1 H+mouse
M1 3 sp.
DLPFC 4 sp.
DLPFC H+chimp.
DLPFC H+marm.
DLPFC H+mac., own table
Lung 3 sp.
Tumour H+mouse
Muscle H+mouse
0.7
0.8
0.9
label transfer across species
scVI, same object
Chimera, kept seeds
(e)